Anand Bhattad

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Education

University of Illinois, Urbana-Champaign (UIUC), USA

Ph.D., Computer Science — *Jan'19 - Present*
Advised by Prof. David Forsyth

M.S., Computer Science — *Aug'17 - Dec'18*
M.S., Civil Engineering — *Aug'15 - Aug'17*

National Institute of Technology Karnataka (NITK) Surathkal, India

B.Tech., Civil Engineering — *July'11 - May'15*

Under Review

4. **Denoising Upsampling Residual Autoencoders: Easy MAP Estimate for Image Enhancement**
   A. Bansal, **A. Bhattad**, D. A. Forsyth
3. **Enriching StyleGAN with Illumination Physics**
   A. Bhattad, D. A. Forsyth
2. **SIRfyN: Single Image Relighting from your Neighbors**
   D. A. Forsyth, **A. Bhattad**, P. Asthana, Y. Zhong, Y. Wang
1. **Image-based Object Insertion using Persistent and Transient Decomposition**
   A. Bhattad, B.Chen, S. Richter, D. A. Forsyth

Conference Papers

5. **Cut-and-Paste Object Insertion by Enabling Deep Image Prior for Reshading**
   A. Bhattad, D. A. Forsyth
   International Conference on 3D Vision (3DV), 2022
4. **DIVeR: Real-time and Accurate Neural Radiance Fields with Deterministic Integration for Volume Rendering (Best Paper Finalist)**
   L. Wu J. Y. Lee **A. Bhattad** Y. Wang D. A. Forsyth
   Computer Vision and Pattern Recognition (CVPR), 2022
3. **View Generalization for Single Image Textured 3D Models**
   A. Bhattad, A. Dundar, G. Liu, A. Tao, B. Catanzaro
   Computer Vision and Pattern Recognition (CVPR), 2021
2. **Unrestricted Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   International Conference on Learning Representations (ICLR) 2020
1. **Improved Style Transfer with Calibrated Metrics**
   M. Yeh*, S. Tang*, **A. Bhattad**, C. Zou, D. A. Forsyth (* for equal contribution)
   Winter Conference on Applications of Computer Vision (WACV) 2020

Workshop Papers

2. **Big but Imperceptible Adversarial Perturbations via Semantic Manipulation**
   A. Bhattad*, M. J. Chong*, K. Liang, B. Li, D. A. Forsyth (* for equal contribution)
   Computer Vision and Pattern Recognition (CVPR) Workshop 2019, ***Oral Presentation***
1. **Detecting Anomalous Faces with "No Peeking" Autoencoders**
   A. Bhattad, J. Rock, D. A. Forsyth
   Computer Vision and Pattern Recognition (CVPR) Workshop, 2018, ***Oral Presentation***

Students Mentoring

- Liwen Wu (2021-present) | Paper at CVPR 2022 | Current Position, PhD Student at UCSD
- Brian Chen (2020-present) | Paper under review | REU at CMU's Robotics Institute
- Kexuan (Klaus) Zou (2019-2020) | Current Position: Software Engineer at NVIDIA
- Anchu Zhu (2018-2019) | Current Position: MS in CS at USC

Research Internships

Intel ISL, Santa Clara/Champaign (remote); *Research Intern* *Feb'21 - Aug'21*
With Stephan Richter and Vladlen Koltun
Project on image-based object insertion.

NVIDIA ADLR, Santa Clara/Champaign (remote); *Research Intern* *May'20 - Dec'20*
With Aysegul Dundar, Andrew Tao and Bryan Catanzaro
Project on single image textured 3D inference via differentiable rendering (CP#3)

Fyusion Inc, San Francisco; *Research Intern* *May'19 - Aug'19*
With Abhishek Kar and Rodrigo Ortiz Cayon
Project on image-based relighting of cars.

Siemens Corporate Technology, Princeton; *Computer Vision & DL Intern* *May'17 - Aug'17*
With Jan Ernst and Ziyan Wu
Developed a semi-automated segmentation tool based on Polygon-RNN from CVPR 2017.

Computational Sensor Lab, Michigan State University; *Research Intern* *May'14 - July'14*
With Prof. Nizar Lajnef and Prof. Amir H. Alavi
Built a framework for Damage Detection Using Self-Powered Wireless Sensors

Earthquake Engineering Research Center, IIIT Hyderabad; *Research Intern* *Jun'13 - July'13*
With Prof. Ramacharla Pradeep Kumar
Designed Earthquake Resistant Multi-Storied Building

Teaching Experience

Computer Science, UIUC; *Graduate Teaching Assistant* *Jan'16 - Present*

- Applied Machine Learning (CS498 AML) in Fall'18 | Designed Kaggle competitions
- Data Structures (CS 225) in Spring'17 | Developed Assignments, Exams & Labs
- Introduction to Computing (CS 101) in Spring'16 & Fall'16 | **Rated as Excellent TA**

Other Projects

Impact of Graph Convolutional Network (GCN) for Active Learning *Mar'18 - May'18*
Project Mentor: Prof. ChengXiang Zhai `[Video | Report | Code]`

Pedestrian Detection and Automated Braking for Self-Driving Cars *Jan'17 - May'17*
Project Mentor: Prof. David A. Forsyth `[Video | Code]`

Services & Leadership

Organizer, External Speaker Series and Vision Lunch Jan 2021-Present

- Hosting the first external speaker series, inviting researchers outside of UIUC for technical talks
- Organizing computer vision reading and presentation group in the CS department of UIUC

Reviewer, Computer Vision Conferences Nov 2018-Present

- Serving as reviewer for CVPR, NeurIPS, ECCV and ICCV conferences and workshops.

Program Committee Jun 2019

- Adversarial Machine Learning in Real-World Computer Vision Systems, CVPR 2019 Workshop
- Security and Privacy in Machine Learning, ICML 2019 Workshop

Organizer, Vision Coffee, Urbana-Champaign *May'18 - Aug'18*

- Vision and Deep Learning paper/topic weekly discussion group
- Finalized agenda/topic, location and initiated discussions

Co-Founder, SchoolEngg *Apr'13 - May'15*

- Forum for high-school students; providing the firsthand knowledge of all engineering disciplines
- Currently, renamed as *PrepLift*, an education counselling start-up in India

Co-Founder, American Society of Civil Engineers (ASCE) NITK Chapter *Apr'13 - May'15*

- Co-Founded the first ASCE student chapter of India
- Raised membership from 11 to 96 in a span of 6 months
- Coordinated with board members, faculty, and outside organizations to facilitate 10+ events